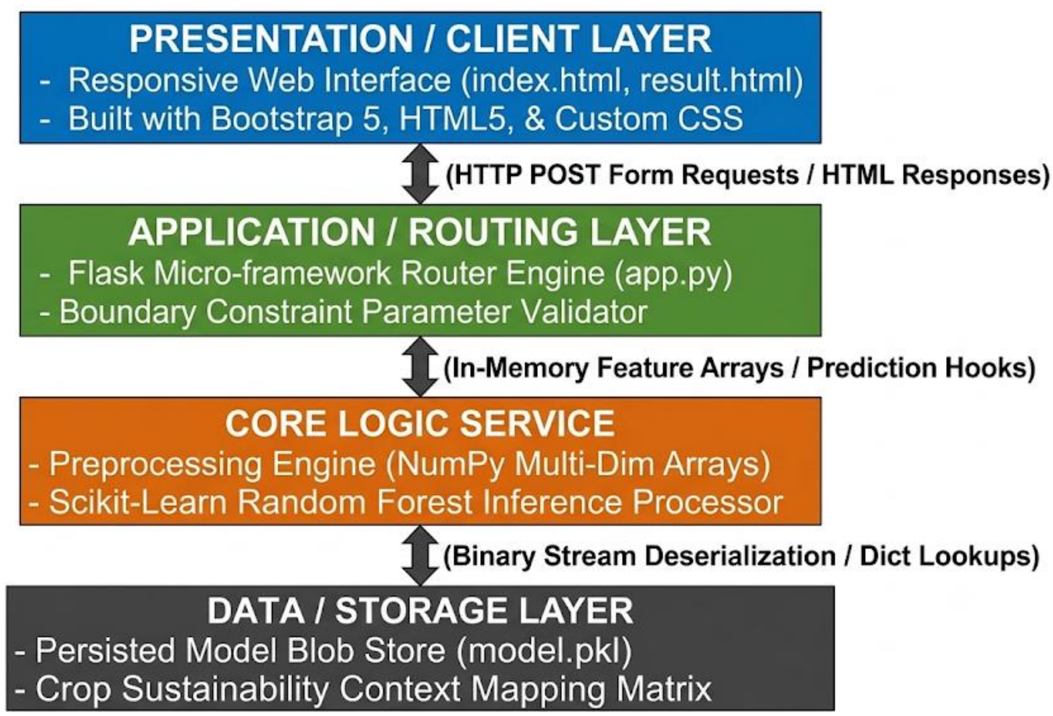
Solution Architecture

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:
Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Serves as the interactive web control console for the end user. It displays input parameters for the seven agro-ecological variables and utilizes a dynamic, tabbed navigation matrix to partition and trigger the three separate user product scenarios cleanly.	HTML5, CSS3, Bootstrap 5 CDN
API / Application Routing	Intercepts HTTP form transmission signals arriving from the web browser client. It validates data limits, handles error handling, routes variable arrays safely to the prediction backend, and dynamically renders the context injection output files.	Flask Framework Server Core (app.py), Python Native HTTP Routes
Core Logic Service	Standardizes and packages raw string inputs into mathematical floats. It structures features into multidimensional matrices matching model dimensions and executes the tree traversal pathways of the predictive algorithm.	NumPy, Pandas, Scikit-Learn Inference APIs
Data / Storage Layer	Holds the compiled algorithmic memory structure and persistent metadata configuration profiles. It loads model trees instantly into RAM via stream deserialization at boot and serves as a local dictionary for regional crop sustainability roadmaps.	Python Pickle Binary Store (model.pkl), In-Memory Context Python Dictionaries